

# Techies Against Facebook: Understanding Negative Sentiment Toward Facebook via User Generated Content

Abu Saleh Md Noman, Sanchari Das, Sameer Patil

School of Informatics, Computing, and Engineering

Indiana University Bloomington

Bloomington, Indiana, USA

{amdnoman,sancdas,patil}@indiana.edu

## ABSTRACT

Researchers have recognized the need to pay attention to negative aspects and non-use of social media services to uncover usage barriers and surface shortcomings of these systems. We contribute to these efforts by analyzing comments on posts related to Facebook on two blogs with a technically savvy readership: Slashdot and Schneier on Security. Our analysis indicates that technically savvy individuals exhibit notably large negative sentiment toward Facebook with nearly 45% of the 3,000 reader comments we coded expressing such views. Qualitative coding revealed *Privacy and Security*, *User Experience*, and *Personal Disposition* as key factors underlying the negative views. Our findings suggest that negative sentiment is an explicit higher level factor driving non-use practices. Further, we confirm several non-use practices reported in the literature and identify additional aspects connected to recent technological and societal developments. Our results demonstrate that analysis of user generated content can be useful for surfacing usage practices on a large scale.

## CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms**; **Social media**; *Collaborative and social computing*;

## KEYWORDS

Facebook, social media, social networking services, non-use, user experience, negative sentiment, privacy, user generated content, technically savvy users

## ACM Reference Format:

Abu Saleh Md Noman, Sanchari Das, Sameer Patil. 2019. Techies Against Facebook: Understanding Negative Sentiment Toward Facebook via User Generated Content. In *CHI Conference on Human Factors in Computing Systems Proceedings (CHI 2019)*, May 4–9, 2019, Glasgow, Scotland UK. ACM, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3290605.3300698>

## 1 INTRODUCTION

Social media services, such as Facebook, have become an integral part of everyday lives of people across the world. These services provide a number of individual and collective benefits [22, 35], despite challenges such as privacy concerns [1], context collapse [65], cyberbullying [38], etc. However, as the platforms and their usage continue to evolve, there are indications that the downsides are having a growing and noticeable impact, leading some people to curtail or abandon the use of these services. For instance, the constant stream of scandals revealing undesirable business practices of the service providers as well as the rise in malicious actors and harmful content on the platforms has created broad calls for stopping the use of these platforms, such as the trending hashtag #DELETEFACEBOOK [29].

While extensive research attention has been devoted to the adoption, use, and benefits of social media services in general, and Facebook in particular [33], relatively few studies have focused on those who express negative views and exhibit various types of disengagement, such as non-adoption, usage reduction, or abandonment [67]. Existing work that examines these aspects is mostly limited to surveys of samples drawn from a general population [10, 49]. Although these studies have yielded important and useful insight, they suffer from two notable limitations. First, the average person in these samples may not possess adequate technical knowledge to understand and notice the underlying operational

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](mailto:permissions@acm.org).  
CHI 2019, May 4–9, 2019, Glasgow, Scotland UK

© 2019 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-5970-2/19/05...\$15.00

<https://doi.org/10.1145/3290605.3300698>

mechanisms of social media services that are often a major contributing factor to the negative perceptions of various features such as behavioral tracking, personalized advertising, third party data sharing, etc. Second, participants in survey studies may refrain from expressing negative opinions to the fullest extent as they might when sharing their unsolicited views in a more naturalistic setting.

To address these shortcomings, we investigated negative sentiment toward social media services by analyzing user generated content produced by technically savvy individuals. We characterized technically savvy persons as those interested in the operational aspects of technology at a reasonable level of detail. Specifically, we address the following research question:

*What factors lead technically savvy individuals to hold negative views toward social media services?*

In tackling the research question, we chose to focus on Facebook since it is the most widely known and used social media platform.

To gather naturalistically-expressed unsolicited views of technically savvy individuals, we utilized reader comments on Facebook-related posts on two online blogs that cover news and developments related to technology: Slashdot<sup>1</sup> and Schneier on Security.<sup>2</sup> Labeling itself as ‘News for Nerds,’ Slashdot posts links and corresponding summaries of online articles of relevance to an audience with a strong interest in technology. The links and summaries are submitted by readers, and each post becomes the topic of a corresponding threaded discussion. Similarly, Schneier on Security is a blog of posts pointing to news, research, or essays on technology topics. Unlike the general technology coverage of Slashdot, posts on Schneier on Security are limited to matters pertaining to privacy and security. In contrast to the reader-submitted posts of Slashdot, each post on Schneier on Security is created and curated by the renowned security expert Bruce Schneier.

Apart from their relevance for our research topic and targeted population, we chose the two sites because each has a long history that predates the inception of Facebook and a *community* of engaged regular readers who contribute comments to the online discussions related to the posts. Moreover, both sites moderate reader comments, thus ensuring quality despite the naturalistic nature of the content [26]. In fact, prior work at CHI has generated useful insight by studying the Slashdot community [39, 41]. We included Schneier on Security due to the blog’s focus on privacy and security which often feature as prominent aspects in research and news regarding Facebook. Moreover, the additional source

allowed us to confirm that our insight applies more broadly beyond a single site.

We used qualitative techniques to analyze 3,000 randomly selected reader comments on posts related to Facebook on the two chosen sites (2,000 comments from Slashdot and 1,000 from Schneier on Security). Our analysis suggests that technically knowledgeable individuals harbor notably high levels of negative views toward Facebook. These views often lead these individuals to engage in various forms of non-use and to urge others to do the same. Our findings confirm several results from past research on Facebook non-use and reveal a few new aspects that could be attributed to recent societal and technical developments regarding the Facebook platform.

The contribution of our work is multifaceted:

- We broaden the discourse on Facebook non-use to consider negative sentiment as an explicit higher level aspect connected to non-use practices.
- We confirm that several of the Facebook non-use practices reported in the literature continue to apply despite the changing nature and user experience of the platform. In fact, we identify additional factors connected to recent technological and societal developments.
- We target a technically savvy population and show that these individuals exhibit large amounts of negative sentiment toward Facebook.
- We show that analyzing high quality user generated content could serve as a useful method for surfacing user sentiment and studying user behaviors and experiences on a large scale.

In the sections that follow, we first situate our work in the literature. Next, we describe how we collected, selected, and coded the user generated content on the chosen sites. We proceed to present the findings of our analyses followed by a discussion that unpacks the insight. We further highlight practical implications of the insight and point out important limitations. At the end, we suggest a few future directions for extending our work and offer concluding remarks.

## 2 RELATED WORK

Prior work related to our research falls under three broad themes: non-use of technology in general and Facebook in particular, negative consequences associated with the use of Facebook, and studies of user generated content, particularly Slashdot. In the subsections that follow, we discuss the salient literature under each of these themes, respectively.

<sup>1</sup><https://slashdot.org/>

<sup>2</sup><https://www.schneier.com/>

## Non-use of Technology and Facebook

Noting that use and non-use are not a pure binary distinction, researchers have been studying practices pertaining to various forms of technology non-use, such as non-adoption, passive use, resistance, or abandonment. For instance, more than 30 years ago, Grudin [25] pointed out that non-adoption and failure of a collaborative system are driven by a misalignment between the burdens and benefits of use. For understanding non-acceptance, non-adoption, and non-use of enterprise and collaboration systems in particular, researchers have provided theoretical insight (e.g., Unified Theory of Acceptance and Use of Technology [63], the Technology Acceptance Models [62], etc.) as well as field studies of specific technologies, such as groupware [50], local area networks [14], etc. Literature in the field of Information Systems (IS) includes a number of conceptual and empirical models and frameworks that characterize resistance toward the use of enterprise systems (e.g., [34, 42, 46], etc.). On a more general level, researchers have attempted to uncover the various socioeconomic and technical factors that underlie non-use of Information and Communication Technologies (ICTs) and unpack their connections to larger societal issues such as the digital divide, technological literacy, etc. (e.g., [59, 60, 64, 67] etc.). Some of these efforts focus on non-use practices of specific populations, such as the youth [18].

Recently, there have been active efforts within the Human Computer Interaction (HCI) community to pay increased attention to non-use [8, 11, 12]. In this regard, Satchell and Dourish [57] argue that HCI studies focus on “use” because it is more visible and describe scenarios where studying non-use might be useful. Baumer et al. [11, 12] stress that non-use can take a variety of forms. For instance, Satchell and Dourish [57] characterized various types of non-use, such as lagging adoption, active resistance, disenchantment, disenfranchisement, displacement, and disinterest.

Specifically, researchers have been paying increasing attention to the non-use of social media. While some of these studies cover non-use of social media services such as Twitter [58], MySpace [27], etc., the bulk of the work in this space is focused on Facebook, the most widely used social media platform. Some researchers have focused on understanding the differences between Facebook users and non-users in terms of individual traits, such as personality [15, 45, 56], finding that Facebook non-users are less shy, less overtly narcissistic, and less sociable and sociotropic while active Facebook users are less conscientious and less lonely. These differences in individual characteristics have been found to lead to further differences in usage patterns and preferences for specific features of the Facebook platform [56] and levels of information disclosure [15]. In a similar vein, Lampe et al. [40] noted that those with higher perceived levels of

bonding social capital are less likely to use Facebook, and Baumer [9] uncovered that Facebook non-use is influenced by socioeconomic inequalities connected to demographic factors, such as age, gender, ethnicity, employment, and income. Wyche and Baumer’s [68] study of Facebook non-users in rural Zambia suggests that the impact of socioeconomic inequalities can be more severe in non-affluent countries, owing to deficient infrastructure, low English literacy, high access costs, etc.

Other studies have uncovered specific non-use practices, such as leaving or limiting [10] or taking long breaks akin to a “vacation” from Facebook [55]. Despite such conscious non-use efforts many of these individuals have reported experiencing “Facebook withdrawal symptoms” and have reverted back to using Facebook [10, 13]. The extent and nature of Facebook non-use as well as the subsequent reversion have been attributed to a number of factors, such as privacy concerns, quality and relevance of Facebook content, work productivity, external influences from friends and family, context collapse, channel effects, etc. [10, 40, 55, 65].

Existing research examining Facebook non-use practices and its drivers has explicitly solicited self-reports from individuals in response to specific questions and prompts via techniques such as interviews or online questionnaires. In contrast, our methodological approach analyzed user generated content providing unsolicited personal views regarding Facebook. Due to its naturalistic, anonymous, and unfiltered nature, such data can yield insight missed by other approaches. Additionally, we focused on a population of technically savvy individuals, unlike most prior research that has not explicitly targeted technically knowledgeable people even when focusing on specific populations, such as students, youth, citizens of developing countries, etc. Technical understanding of the operational details of the Facebook platform, such as access control, encryption, Web browser cookies, etc., can serve to highlight problematic aspects, especially regarding privacy and security. In fact, prior work in the privacy and security domains has shown that those with technical expertise are notably different in terms of system understanding and use [23, 30]. Since we did not have direct access to privacy and security experts, we chose to examine the non-use practices of technically savvy individuals, who are typically enthusiastic and early adopters of technology and can thus provide equally instructive insight.

## Negative Consequences of Facebook Use

Research on the use of Facebook typically highlights the various individual and societal benefits of the use of the platform. For example, Facebook use has been linked to a variety of benefits such as acquiring social capital [22], increasing well-being [48], finding employment [16], receiving social support during stressful events [3], and so on.

However, as the adoption and use of the Facebook platform has continued to grow and evolve, researchers have increasingly started to notice several negative consequences associated with the use of Facebook. Apart from privacy concerns that have been an ongoing issue from the early days of Facebook [1], more recent studies have linked Facebook use to other negative aspects, such as envy (e.g., [36]), depression (e.g., [5]), cyberstalking (e.g., [19]), cyberbullying (e.g., [38]), etc. In fact, Islam and Patil [31] suggest that there may be a threshold beyond which the benefits derived from Facebook use begin to reverse, and Wisniewski et al. [66] show that an inability to achieve desired levels of privacy reduces the potential benefits of using the Facebook service.

In the past couple of years, the Facebook platform has been implicated in the spread of targeted misinformation, especially pertaining to important political events, such as the 2016 Presidential election in the US [4] and the Brexit referendum vote in the UK [17]. Scandals regarding Facebook's handling and sharing of user data have been a steady feature in news stories throughout 2018 [43], leading the US Congress to invite testimony on these matters from Facebook CEO Mark Zuckerberg [44]. Owing to these developments, the general public seems to be becoming increasingly aware of the negative aspects of using the Facebook platform [29].

A large proportion of the studies pertaining to Facebook non-use were conducted prior to the above events. As a result, the impact of the negative sentiment generated by these events on non-use of Facebook has not yet been studied. Our analysis addresses this gap by examining user comments posted as late as February 2018. Moreover, the Facebook platform as well as the demographic characteristics and technical literacy of its user base are continually evolving. It is important to examine whether this evolution has served to address and mitigate the previously reported factors that lead to Facebook non-use. To that end, we compare our findings with those from previous studies of Facebook non-use.

### Studies of User Generated Content

Nowadays, many systems allow and encourage their users to contribute content in a variety of forms, such as comments, reviews, ratings, etc. Open-ended text comments in particular can help drive online discussion and discourse and contribute to fostering a sense of community within an online space [53]. As such, naturalistic user comments within an online community can provide insight regarding the sentiment and practices pertaining to the topical domain of the community [28]. For instance, researchers have generated useful findings by studying discussions within online communities devoted to various matters that require social support (e.g., [52, 54]). Therefore, we chose to collect data from Slashdot and Schneier on Security, two online communities that involve our target population of technically savvy

individuals discussing topics connected to the focus of our research, Facebook.

Of our two data sources, Slashdot has been the subject of a variety of research endeavors. Kunegis et al. [37] employed network analysis techniques to study user-provided relationship tags and link level characteristics among individuals in the Slashdot community. In terms of the content, researchers have found that Slashdot discussion threads exhibit strong heterogeneity and self-similarity throughout the different nesting levels of the conversations. The discourse on Slashdot has been characterized as a public sphere that provides a model for idealized democratic debate [7, 51] and exhibits a commitment to open discussion [26]. Halavais [26] described the heavy technological orientation of Slashdot topics and readers, and Lampe et al. [39, 41] showed that the sociotechnical features of the moderation and filtering mechanisms on Slashdot facilitate effective filtering of low-quality comments. Therefore, we believe that high quality user comments from the Slashdot community on posts pertaining to Facebook are well-suited for our investigation. While Schneier on Security has not received similar research attention, it shares several of Slashdot's characteristics, such as a technically savvy target audience, moderated comments, and open discussion. Unlike the general coverage of technology on Slashdot, Schneier on Security focuses exclusively on privacy and security matters, which can provide a useful topical focus given that privacy and security concerns are prominently mentioned in relation to the Facebook platform.

## 3 METHOD

As mentioned earlier, we collected all Facebook-related posts and associated reader comments from two sites: Slashdot and Schneier on Security. The posts covered the period from Facebook's inception in 2004 until the end date of our data collection in February 2018. We randomly selected a sample of these comments for in-depth qualitative analysis. In the following subsections, we describe our data collection, comment selection, and coding procedures.

### Data Collection

For both selected data sources, we scraped the respective sites for all posts related to Facebook since Facebook's emergence in 2004 until the end of our data collection in February 2018 for Slashdot and November 2017 for Schneier on Security. We also collected all reader comments associated with the scraped posts.

*Slashdot.* Each post on Slashdot is associated with a topic tag, such as *Political*, *Facebook*, *Social media*, *Science*, etc. We collected 940 Slashdot posts with the topic tag 'Facebook' along with 78,597 reader comments associated with these 940 posts. Reader comments on a Slashdot post are displayed as

a threaded discussion. The comments are moderated by the readers of the site to ensure content quality and to flag spam or other undesirable content. Each moderator on Slashdot is able to modify the score of a given comment by  $\pm 1$ , and each comment has a cumulative score between -1 to +5. Further, each comment is classified into one of several categories: *Interesting*, *Insightful*, *Informative*, and *Funny* as the positive categories and *Flamebait*, *Troll*, *Off-Topic*, and *Redundant* as the negative categories. This extensive and informative moderation environment makes Slashdot content a rich source of high quality user generated content [26, 39, 41].

*Schneier on Security*. All posts on Schneier on Security are written by the renowned security expert Bruce Schneier. Like Slashdot, each post is tagged with one or more tags relevant to the content of the post, such as *Algorithms*, *Identification*, *Privacy*, *Drones*, etc. Readers of the blog can leave comments without needing to sign up for an account on the site. The comments are moderated, thus ensuring that off-topic or spam comments are rare. We found that selecting posts with the tags ‘Facebook,’ and ‘Social Media’ yielded only 69 posts since 2004. On the other hand, selecting posts based on the presence of these two terms in the title or content of the posts returned a large number of posts irrelevant to our research. As a result, we manually examined every Schneier on Security blog post from 2004 onward and selected 237 posts that were marked by three independent coders with 100% agreement as pertaining to our research question regarding Facebook. We then collected the 11,365 reader comments associated with these 237 posts.

### Comment Selection

From the combined set of 1,177 posts and 89,962 associated individual comments, we first filtered out comments that were less than 100 characters in length and those that did not contain either of the terms ‘Facebook,’ or ‘FB.’ For Slashdot comments, we further narrowed the set to include only those comments with a moderation score of +2 or higher. These criteria ensured that we removed comments without enough content to enable understanding the text as well as those that were marked as ‘low quality’ by the Slashdot moderation processes. From this filtered set of comments, we randomly selected 3,000 comments for deeper examination and analysis, 2,000 from Slashdot and 1,000 from Schneier on Security. These 3,000 comments cover a period spanning 2006–2018 (2010–2018 for Slashdot and 2006–2017 for Schneier on Security).

### Coding

We performed qualitative coding on the set of 3,000 comments in two stages:

*Sentiment Coding*. The first and the second author independently marked whether a comment expressed negative sentiment toward Facebook, such as criticism, frustration, annoyance, etc. The two coders initially coded 500 comments from Slashdot, achieving 78.3% agreement with Cohen’s  $\kappa = 0.56$ . Next, the coders discussed and resolved 20 of the discrepancies and revisited the initial coding in light of the discussion. The revised coding reached an intercoder agreement of 92.8% with Cohen’s  $\kappa = 0.856$ . Given the high agreement achieved for the initial 500 comments, the two coders then independently coded the remaining 2,500 comments, achieving an overall agreement rate of 93.6% with Cohen’s  $\kappa = 0.872$ . The coding revealed that negative sentiment toward Facebook was expressed in 1,342 of the comments, 1,022 from Slashdot and 320 from Schneier on Security.

*Thematic Coding*. Next, we utilized techniques from Grounded Theory [21, 24] to engage in an inductive and comprehensive qualitative thematic analysis [32] of the 1,342 comments identified in the previous stage as expressing negative sentiment toward Facebook. The first and the second authors generated codes via initial open coding. Next, the two coders used axial and selective coding to identify relationships and thematic groupings among the codes. Table 1 shows the 10 themes that emerged from the coding process.

Subsequently, the two coders independently categorized an initial set of 100 comments with negative sentiment toward Facebook into the identified themes. It should be noted that the themes are not mutually exclusive; i.e., a comment may be coded as belonging to more than one theme. For example, consider the comment: “*Maybe I’ll see no ads after changing my Facebook age to something ridiculous? Why does anybody put their real age/birthday into Facebook?*” This comment was coded under the themes *Advertisements* and *Privacy and Security*. Given that negative sentiment can be a strong indicator of non-use, the coders further marked whether the comment explicitly mentioned that the writer engaged in some form of Facebook non-use, such as non-adoption, reduced usage, lurking, self-censorship, etc. For the initial set of 100 comments, the two coders reached an agreement of 89.8% with Cohen’s  $\kappa = 0.87$ . Since the initial agreement was high, the coders proceeded to code the remaining 1,242 comments with negative sentiment toward Facebook, reaching an overall agreement of 97.3% with Cohen’s  $\kappa = 0.966$ .

## 4 FINDINGS

At the end of the sentiment coding described in the previous section, the two coders agreed on classifying 1,185 out of the sample of 3,000 comments as containing negative sentiment toward Facebook (888 for Slashdot and 297 for Schneier on Security). The two coders further agreed that 1,658 comments

did not express negative views regarding Facebook (978 for Slashdot and 680 for Schneier on Security). In the case of the remaining 157 comments, only one of the coders classified them as containing negative sentiment toward Facebook. Given the high level of agreement between the two coders ( $2,843/3,000 = 94.8\%$ ), we decided to mark these 157 comments as containing negative sentiment because at least one of the coders classified it as such.

As described in the previous Section, we engaged in thematic coding of these 1,342 comments (1,022 for Slashdot and 320 for Schneier on Security). The results of the coding are summarized in Table 1. Apart from longer length along with an expected privacy and security bent of the comments on Schneier on Security, we did not find any notable differences in the comments from the two data sources. The following subsections describe in detail each of the 10 themes that emerged during the thematic coding of the combined comment set followed by a subsection on explicit mentions of any form of non-use.

### Privacy and Security

Unsurprisingly, privacy and security featured prominently among the expressed concerns. The amount of surveillance and third party sharing of online activities and communications and related metadata was worrying for many individuals: *“...If Facebook can do it, the NSA [United States National Security Agency] can do it, too. In a few years, everybody can do it. This is one of the reasons why I don’t have pictures of myself on Facebook.”*

Invasion of privacy due to data collection and mining and information sharing with third parties was found to be an undesired aspect even when it was used to offer features that could potentially serve a useful purpose: *“Facebook is actively developing new, innovative ways to invade your privacy, and this particular bit of data mining technology has become reliable enough that they felt it would be good PR [Public Relations] to create a feel-good, help-the-disabled feature out of it.”*

A perceived lack of care regarding user privacy and security concerns from Facebook as well as other technology companies served to strengthen negative views: *“...Of course, Facebook isn’t the only company guilty of this type of thing – and I suspect that until there is some serious consequence associated with this type of security hole, most companies won’t take it seriously enough.”* A number of comments argued that Facebook is exploiting its monopoly in the social media space and prioritizing profits over better service and experience for the users.

Commenters were displeased that Facebook’s Terms Of Service (TOS) require the use of their real names: *“Not to mention, Facebook’s TOS is that you must use your real name when creating an account. If FB cared one whit about privacy it*

*would let people use aliases. It really is incredible how ghetto and scammy FB is with their tactics and policies.”* Commenters noted that these terms increase the ease and likelihood of identity theft: *“By signing up for Facebook, you surrender your identity as it is...”* Some commenters reported that they consciously violate these terms to avoid the associated privacy and security consequences: *“I have a fake account of Facebook, it’s useful when researching. Considering I use it as an information resource and since I value my privacy, there is no way I would put my personal information on there! Muffin got it right. Don’t use it.”* Relatedly, a commenter suggested being open to a ‘business account’ with access and use restricted within a work environment: *“...Never had an account on FB... never will... I don’t want to give any personal info to them, but I would consider a business only account if they offered one.”*

Our technically savvy commenters argued for open source code as a potential solution for some of the privacy and security concerns: *“If Facebook source code was open, security flaws which are disclosed every one day or another, probably ruining life of thousands due to privacy leaks, would have been closed faster...”* In this regard, it was pointed out that Facebook promotes ‘openness’ as one of its core principles yet reveals little about its own operations and algorithms: *“So if G [Google] and FB are OK with privacy being dead, why don’t they share the details of their advertising operations and source code? Or do they mean privacy is dead for everyone but themselves?”*

### User Experience

A number of aspects related to the Facebook user experience were perceived negatively, such as bandwidth consumption, auto-playing videos, etc. One unhappy commenter remarked, *“Not a fan of this idea. [injecting advertisements within chat threads] Facebook Messenger spams could exceed the data caps on mobile plans (\$\$\$) and create a backlash from angry customers.”* Others complained about the user interface: *“I’m surprised no one mentioned the disgusting interface Facebook has. It really seems badly made and unsophisticated...”*

These various user experience issues impacted people’s use of Facebook. As one person remarked: *“If Facebook continues to make its site user-unfriendly, I’ll simply stop using Facebook. I’ve already dropped back on my usage because I cannot view my timeline the way I want to view it...”* User experience features connected to the ability to control the appearance of the profile and to target information disclosure to appropriate parties were noted as particularly important. Problems with these aspects led to great frustration: *“Just went into my profile to try to remove / disable this POS and you are not even given the option to do so... I am so close to closing my Facebook account it is not even funny anymore.”*

Some of the negatives pertaining to user experience were based on how Facebook handles the actions of other people

**Table 1: Results of thematic coding showing the comments with negative sentiment toward Facebook falling under each theme for each of the two data sources along with the fraction of those comments that explicitly indicated Facebook non-use. Percentages for each column do not add to 100% because the themes are not mutually exclusive.**

| Themes                  | #Comments (Slashdot) |                  | #Comments (Schneier on Security) |                  |
|-------------------------|----------------------|------------------|----------------------------------|------------------|
|                         | Overall              | Explicit Non-use | Overall                          | Explicit Non-use |
| Privacy and Security    | 432 (42.27%)         | 138 (44.37%)     | 212 (66.25%)                     | 111 (68.94%)     |
| User Experience         | 341 (33.36%)         | 118 (37.94%)     | 41 (12.81%)                      | 18 (11.18%)      |
| Advertisements          | 104 (10.17%)         | 21 (6.75%)       | 14 (4.37%)                       | 5 (3.1%)         |
| Psychosocial Well-being | 9 (0.88%)            | 1 (0.32%)        | 7 (2.18%)                        | 3 (1.86%)        |
| Personal Disposition    | 147 (14.38%)         | 42 (13.5%)       | 94 (29.37%)                      | 68 (21.86%)      |
| Alternate Options       | 18 (1.76%)           | 8 (2.57%)        | 18 (5.62%)                       | 10 (6.21%)       |
| Uninteresting Content   | 39 (3.81%)           | 8 (2.57%)        | 28 (8.75%)                       | 12 (7.45%)       |
| Politics                | 30 (2.93%)           | 5 (0.01%)        | 1 (0.31%)                        | 0 (0%)           |
| Fake accounts and bots  | 5 (0.49%)            | 2 (0.64%)        | 22 (6.87%)                       | 8 (4.96%)        |
| Miscellaneous           | 35 (3.42%)           | 11 (3.53%)       | 34 (10.62%)                      | 14 (8.69%)       |

on the platform. For instance, one commenter remarked that “An hour on Facebook will make you hate humanity.” In addition, commenters complained about Facebook’s attempts to link people based on their actions: “...I find it quite hard to believe that Facebook matches people who happened to be in the same location for some time. That would generate an awful high number of false positives...” The suggested people links were also criticized for lacking social nuance: “The whole ‘people you may know’ thing is kind of creepy. I once had Facebook suggest a person who had died not long before the friend suggestion.” Commenters further found it concerning that Facebook compiles information on them without consent simply based on the actions of the Facebook users they know: “Scariest, is that you don’t even need to have a Facebook account for them to have a dossier on you. All it takes is countless ‘friends’ uploading their contact list to try to maximize friendship.”

In addition, the user experience of Facebook features designed for developers who build apps for the Facebook platform was relevant for our technically savvy sample: “The Facebook API is a tapestry of bad designs that are not even consistent across the various namespaces. The developer console looks like something that was designed by people who have never heard of UX principles, and the documentation is awful. I’d rather work with Yahoo CSV [Comma Separated Values].”

### Advertisements

Advertising was a specific aspect of user experience that bothered many individuals. Given that Facebook makes money from showing advertisements to its users, some people felt

that users are entitled to a share of the profits: “if Facebook is making \$1 off of my inconvenience, Facebook should pay me at least .50c of that money ...” Despite the dislike for advertising, some commenters argued that users must resign themselves to tolerating it: “After all, Facebook users have NO CHOICE but to use Facebook and allow them to force you to watch ads. Really, you have no choice, none at all. Suck it up and get used to it.” As others remarked, the only alternative is not using Facebook at all: “Maybe it’s time for you to stop using Facebook altogether. Your continued use of the site IS their explicit permission that they can serve you their content – ads and all.” In fact, advertising contributed to people leaving the platform: “...it was an ad that really struck me and contributed to the many things that got me off of Twitter/Facebook and all the other wannabe sites.”

Apart from the annoying User Experience of advertising, many people were concerned about the Privacy and Security intrusions required for personalizing and targeting the delivery of the advertisements: “...I’m sure Facebook has much, much more data on a given user (at least in 2012); they would have to in order to make the targeted ads...right?” Many commenters did not appreciate being profiled based on their interactions and communications on the platform: “...Facebook probably mines the unencrypted messages to help form an ‘advertising profile’ for you so they can better target ads at you when you’re on Facebook.” In addition, commenters noted Facebook’s broad collection of information by tracking individuals across the Web via cookies and the ‘Like’ button and worried about security threats from advertisements from malicious actors.

## Psychosocial Well-being

Facebook was accused of creating a negative impact on individual and social well-being: *“The negative impact social media has on the human psyche is, in my opinion, quite significant. FOMO [Fear Of Missing Out], self-esteem issues and F4ceb00k depression are real things and they exist with a measurable amount of people who live through mass social media...”* Some people left Facebook owing to these issues: *“... Quitting Facebook was one of the best decisions that I ever made. Nothing good comes from it. It encourages us to compare ourselves negatively to others whom may have more money or more success. Facebook is psychologically damaging.”*

People found this aspect even more problematic in the cases of those who were already suffering from mental health issues prior to becoming a Facebook user: *“The problem with social media (not just FB) and depression is not that people do nothing but stare at FB and get depressed. The problem (or, one of them) is that if you’re already depressed, viewing social media can make it worse...”* Some pointed to Facebook having a negative impact on family life: *“... If your kids did not grow up in the age of Facebook, then it is uncharted territory to you. You don’t know the dangers, the addiction, the toll it takes on a family as they drop all activities and everything, for Facebook activity...”* Others were worried about the larger impact on societal well-being by noting that a sizable proportion of the population in the US (*“a pill-popping nation”*) is suffering from anxiety and depression: *“... the masses are unstable and the trend is worsening – Social media correlations with higher rates of reported mental illness...”*

## Personal Disposition

If a comment did not provide sufficient content or context for the expressed negative sentiment, we attributed the sentiment to personal disposition or preferences. The comments in these category include statements such as *“I do not care,” “waste of time,” “do not like using it,”* etc. Many people expressed their negative disposition toward Facebook via swear words or derogatory modifications of the word Facebook: *“you really don’t need to follow the herd to Faceache and Twat-ter.”*

People’s personal disposition toward Facebook was driven by a number of factors, such as a perceived lack of need or utility (*“... Facebook is a waste of time simply because Facebook neuters your ability to reach people.”*) or privacy and security concerns (*“Protect yourself by forcing the authentication to happen over TLS [Transport Layer Security]. Or stop logging in to Facebook from public networks. OR, do the sensible thing and don’t use Facebook at all.”*)

Some commenters felt that Facebook is a fad that will soon be replaced by the ‘next big thing.’ Commenters pointed to

the lack of any negative consequences from not using social media: *“... I did [stop using social media] 4 years ago and I’m surviving just fine. No Twitter, no Facebook, no Instagram. Nothing.”* Some recommended that everyone should follow their example and stop using social media: *“People should delete their Facebook account and other social media accounts.”* In contrast, we also identified comments where people mentioned using Facebook reluctantly due to social or professional reasons, despite their negative personal disposition toward the service: *“It keeps me vaguely in the loop of events and makes me available for my friends and family to contact me. But I keep my privacy options up-to-date and don’t use FB as authentication for anything else. Nor do I use many FB apps or games.”*

## Alternate Options

A number of comments described alternate options for achieving the same purposes without being impacted by the negative aspects of Facebook: *“... There is already an alternative project being created. Does nobody know about Diaspora? It’s open source and distributed, with no central company collecting all your private data... If you have privacy issues with Facebook, spread the word about Diaspora!...”*

Often times, the alternate options covered specific aspects of Facebook functionality, such as broadcasting messages: *“For people with a large audience, there’s Twitter, until Facebook buys it...”* People reported switching to other services for direct communication as well: *“... In any case, I wouldn’t trust anything from Facebook as far as security/privacy goes. I use Signal.”* Some commenters pointed out that direct communication is possible via a variety of means that do not require Facebook: *“But I tell ya what – Facetwat is not on my phone. (yes, Facetwat is my derisive mashup of Facebook and Twitter.)... You wanna reach me? Email me. Text me. iMessage me. Or fucking call me.”*

Apart from their own practices, commenters observed that other populations as well are switching away from Facebook: *“The kids have already moved to Snapchat ... that’s the end of growth and growth is holy in Silly Valley...”*

## Uninteresting Content

Many people’s negative views toward Facebook were driven by their perception of the content on Facebook as being uninteresting, unimportant, or irrelevant for their purposes: *“I see very little content that’s actually from the people I follow ... I hardly ever use Facebook anymore because most of that re-shared content doesn’t interest me anyway.”* Some commenters pointed out that user posts on Facebook are unverified, thus cannot be trusted to provide an accurate and complete picture of reality: *“... If Facebook contact, which is hardly face to face and certainly not person to person, is your social life, then you may have no life at all ... Facebook et al. remind me*



*of something I have observed in the schools of America today, called the 'liars club'...*"

Commenters noted that some of the uninteresting content involves undesirable discourse, such as arguments, abuse, gossip, trolling, etc.: *"I still don't have a Facebook account, and am no worse the wear for it. I have noted that of my family and friends who do have accounts, the ones who typically talk about their Facebook activity the most are definitely the women, and a lot of that talk seems to swing between gossip and outright vicious assaults. I'll just stay out of that mess, thanks."* As another commenter remarked: *"...Facebook these days seems to just be an outrage platform with different sides all screaming at each other and nobody listening..."*

In particular, Facebook was criticized for enabling the spread of disinformation, such as propaganda, hoaxes, conspiracy theories, etc.: *"I wouldn't be surprised if readership is falling. After I had to open a Facebook account last year, it looked interesting for about a week, then it became an annoyance, now it seems to be troll land ... And as quickly as fake news is deleted, new ones will pop up, and will make note of being deleted, which will feed into conspiracies."*

## Politics

Politics was explicitly mentioned as a type of Facebook content that people found inappropriate and undesirable: *"I quickly learned to use the 'mute' feature in Facebook because of those idiots who post zero impact political comments..."* Some commenters questioned the credibility of Facebook as a source of political news: *"Facebook does have credibility? Was that before or after Facebook admitted to promoting pro-Hillary and suppressing pro-Trump stories/outlets?"* Commenters expressed concerns that Facebook can act on behalf of global powers that seek to manipulate and censor online civic discourse.

Some commenters recognized that the content personalization of Facebook can create information bubbles that reinforce existing beliefs and suppress opposing thoughts: *"Yes, Facebook is keeping me in a political bubble, but not nearly to the extent that National Review did in the early 90s ..."* At the same time, commenters also reported being annoyed by what they perceived as overexposure to opposing political views: *"And this is why you don't go on Facebook ever. Well that, and I literally couldn't give less of a damn about the kind of leftist drivel that populates most of any social media."*

Relatedly, there was concern and apprehension regarding the potential use of Facebook data to support the political aspirations of Facebook CEO Mark Zuckerberg: *"...Mr. Zuckerberg is reportedly preparing for a presidential run. The appearance is with his big money and data access he is trying to buy the presidency ... Can Facebook and its executives be trusted?"*

## Fake Accounts and Bots

Commenters pointed to the proliferation of fake profiles on Facebook despite its policy of requiring a profile to be associated with a person's real name: *"the irony I see in this is that if Facebook didn't try to imply that they care about using real names/make it so much of a 'MEATSPACE PROFILE' (they don't, people use pseudonyms/create fake profiles lately), I wonder if Facebook would be as popular or as filled with uselessness as it tends to be."*

Commenters were worried about malicious activities via fake Facebook accounts: *"Since Facebook doesn't verify your identity when signing up for an account ... how long before the bad guys start setting up fake accounts or hacking FB accounts and ransoming people?"* For instance, commenters mentioned that fake accounts were employed by bots to carry out tasks such as promoting and "liking" content or products via automated means, usually for a fee: *"Because I have a hard time believing that 1/7 of the world population is actively on Facebook and I also know full well that there are millions of bot accounts whose likes are purchased."*

The market for purchasing bot-generated Facebook Likes impacted the credibility of Facebook content: *"Technologists and inventors need to work on cutting Facebook off at the knees. Disposable identities, fake searches, encrypted fake traffic, photos of non-existent people, artificial-intelligence tools for generating fake blogs and comments, fake clicks, a tsunami of false information..."* It also eroded trust in Facebook as a company: *"1,000 Facebook likes for \$29.99. The unknowable number of times Facebook has betrayed, bamboozled, lied to, and scammed their own user base makes it quite difficult for me to have even one byte of sympathy for Mr. Z and his shaky marketing company."*

## Miscellaneous

Comments that mentioned any issues apart from the above nine major themes were gathered under a single theme of miscellaneous concerns. For instance, some commenters complained about the Facebook app's excessive use of the smartphone battery: *"Years ago I deleted the Facebook app due to excessive battery drain."*

A number of people were dismayed by the use of Facebook by prospective employers: *"The reason employers want everyone to be on social media: They can use it to gather information about you that would be illegal or inappropriate to ask in a job interview."* In this regard, those who did not use Facebook believed that they might be judged negatively by prospective employers due to a lack of Facebook presence: *"I don't use Facebook ... If an employer wants my years of experience they will take me as I am. If they are going to reject me because I don't waste time on Facebook, then I probably wouldn't last long there. Their loss."*

On rare occasions, commenters expressed their extreme negative sentiments via sarcasm and absurdity: *“Due to the fact that Facebook is nothing more than prostituting your social life in exchange for some IT service it makes sense that Facebook should be rated XXX and that Facebook should be prosecuted for boosting child prostitution if they do not take care that no child gets a Facebook account.”* There was exasperation at being in the minority regarding recognizing the negative impacts of Facebook: *“The overwhelming majority of people use Facebook. They don’t give a shit about the ideals of freedom and decentralization. And they outnumber us hugely. We can’t ‘win’ in an essentially democratic system wherein a tiny ‘we’ is up against a massive horde of ‘them.’”*

### Explicit Non-use

As mentioned earlier, we examined the comments to check whether the expressed negative views were linked to non-use of Facebook. We found that 472 out of the 1,342 comments with negative sentiment explicitly mentioned non-use of Facebook (311 for Slashdot and 161 for Schneier on Security). Among these 472 comments, the top three themes were *Privacy and Security*, *User Experience*, and *Personal Disposition*.

Regarding *Privacy and Security*, we found that non-users were concerned about Facebook’s default privacy and security settings, risk of identity theft, selling or sharing of personal data for advertising purposes, tracking and surveillance of online activities, automatic posting of activities to their Facebook timelines, targeted and personalized advertising, malicious applications, presence of fake profiles, data retention despite deactivating the account, inability to control incoming content, etc. One commenter explained: *“I used to use Facebook since the early days... But then I deleted it... Got sick of the privacy issues, having my personal information being sold for money (while I get no benefit from it)...”*

In terms of *User Experience*, factors that led to non-use included: problematic user interfaces, unsolicited friend suggestions, being tagged in photos by others, unwanted gaming requests, inadequate ability to block unwanted content, irrelevant advertisements, problematic face recognition in photos, etc.: *“... I know a LOT of people that have quit Facebook... Why? because FB has made it an annoying piece of crap. you cant block all stupid game requests by default... Facebook utterly sucks compared to a year ago... It’s fricking turning into MySpace.”*

## 5 DISCUSSION

In contrast to prior research on non-use and negative impacts of Facebook, we focused exclusively on a sample of technically savvy individuals. While we do not know the level of technical knowledge of the individual commenters, we relied on the expected strong self-selection regarding

commenting on the chosen sites oriented toward a technically savvy audience. We found that our technically savvy sample harbored large amounts of negative sentiment toward Facebook, with nearly 45% of the comments in our sample of 3,000 expressing negative views and nearly 35% of these explicitly mentioning non-use of the platform.

These high proportions are striking given that technically savvy individuals are typically early and enthusiastic adopters of any technology. We believe that the high levels of negative sentiment and corresponding non-use can be attributed to our sample’s deeper understanding of operational details that led them to recognize problematic aspects of the platform. Indeed, several of the illustrative comments we reported in the previous section refer to sophisticated technological matters, such as encryption, cookies, data mining, API, source code, etc. Moreover, in contrast to prior work, we did not find evidence of reversion [13], suggesting that the sentiments and convictions of our technically savvy sample may be stronger compared to the average person.

As mentioned earlier, the themes reported in the previous section are neither mutually exclusive nor exhaustive. As several of the supporting comments reported for each theme indicate, a comment could touch upon more than one theme. Moreover, some of the themes are related to each other as well. For instance, advertising could be perceived as a user experience annoyance, a tradeoff between personalization and privacy, or a security threat from malicious content. Therefore, we coded each comment into all applicable themes that were mentioned within the text of the comment.

Some of the early findings on Facebook non-use were reported more than five years ago. Since then, there have been notable changes in the Facebook service, the user experience of the platform, the preferences and practices of user base, people’s awareness and understanding of the system, the laws and regulations pertaining to technology and privacy, etc. Yet, our themes echo results from the literature regarding various forms of Facebook non-use, such as limiting usage or leaving the service, and the causes underlying these practices [6, 10, 40]. For instance, our theme of *Psychosocial Well-being* is consistent with the findings of prior research that linked Facebook use to depression and reduced well-being [47, 61], and the *banality* reported by Baumer et al. [10] falls under our theme of *Uninteresting Content*. Similarly, the comments we analyzed include mentions of what Baumer et al. [10] described as *lagging resistance* where people resort to reluctant use of Facebook due to social or professional considerations.

Our findings thus indicate that the factors that lead to non-use do not yet seem to have been addressed. In fact, we provide additional drivers of non-use connected to recent technological and societal developments, such as “fake news,” political arguments and polarization, personalization

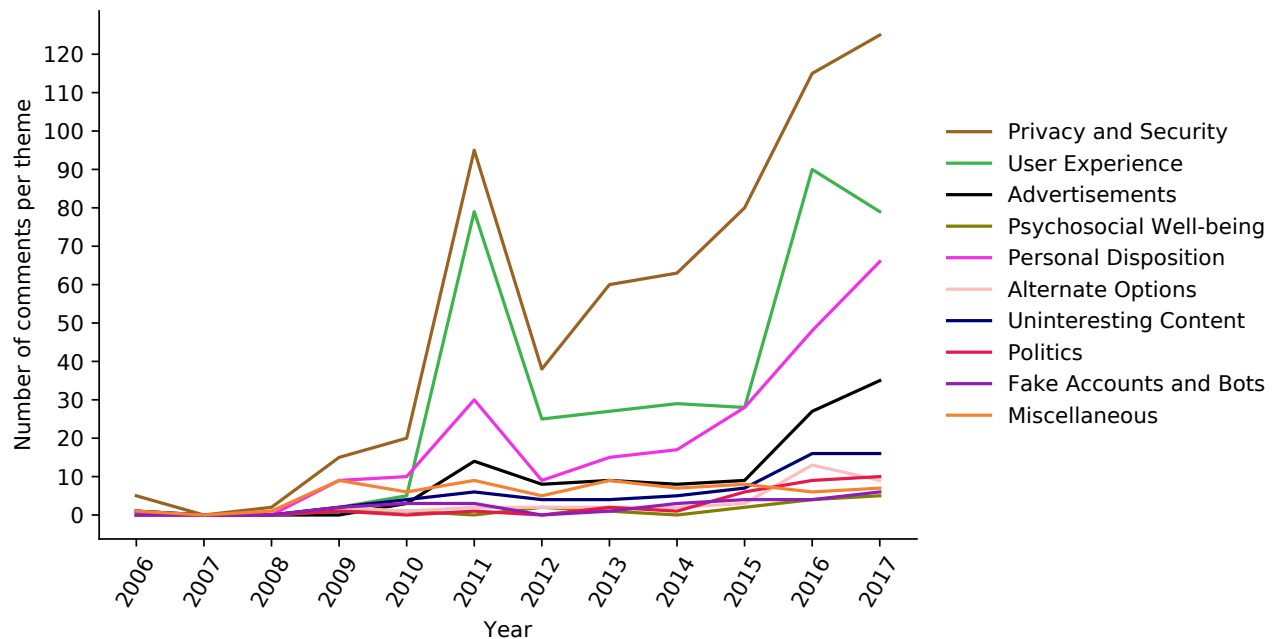


Figure 1: Number of comments per theme by year, combined across the two data sources.

and targeting of promoted content, etc. While interpersonal privacy, such as privacy from friends, family, or employers, was mentioned, we found a far greater mention of consumer privacy and corporate and government surveillance. Taken together these factors seemed to raise privacy and security concerns and degrade the user experience, thus increasing people's proclivity to view Facebook in a negative light and reduce or abandon its use. These trends can be observed in Figure 1 that shows the number of comments from both data sources combined that fell under each theme per year from 2006 until 2017, the last year fully covered by our data. While Figure 1 provides a quick sense of temporal patterns, we should stress that the absolute numbers cannot be taken as statistically meaningful owing to our random sampling and qualitative coding techniques.

Congruent with the Technology Acceptance Model [20, 63], the expressed negative sentiment often indicated that the commenters believe that the perceived risks and costs of Facebook use outweigh the benefits. Notably, we found that negative sentiment often created a desire to influence other parties. For instance, commenters suggested non-use to others and advocated technical or regulatory changes that could address the underlying causes.

## 6 IMPLICATIONS

Each of the themes described in Section 4 covered factors that engendered negative sentiment toward Facebook which

in turn led to non-adoption, limited use, or abandonment of the platform, thus suggesting that negative sentiment could provide a higher level conceptual lens for studying and predicting non-use. Intention and sentiment are strong indicators of behavior [2]. As illustrated by our findings, negative sentiment is thus likely to drive and indicate non-use. Moreover, expressed negative sentiment may strongly *imply* non-use even if it is not explicitly mentioned. In addition, negative sentiment could describe or prescribe non-use by parties other than the one expressing the sentiment. Further, examining negative sentiment can enable a richer and diverse capture of the pulse of the population.

Our findings demonstrate that unsolicited naturalistic user generated content can serve as a source of high quality data for uncovering user sentiment and practices. Our method corroborated past findings on Facebook non-use that were derived mostly via questionnaires and interviews and contributed additional insight beyond those results. Unlike the explicitly solicited data collected via self-reports and laboratory studies, user generated content is voluntarily provided. As such, it can provide unfiltered and strong responses as long as there are adequate moderation mechanisms in place to filter out off-topic and irrelevant comments. Given the increasing and open availability of such content, it has the potential to serve as a useful source of data on user sentiment, preferences, and practices for researchers and system designers alike.

Systems designed for a general population are typically evaluated in terms of their use by the average person without specialized technical knowledge. Our findings underscore the importance of focusing on technically savvy users as well. As our findings show, technically savvy individuals are more likely to recognize negative aspects linked to the operational details of the system. If not addressed, the number of people who feel negatively about these aspects can grow over time as the average user gains more technical knowledge with use and experience. For instance, the increasing coverage of privacy and security related scandals connected to Facebook has led to an increased awareness of these issues among the general public. Moreover, technically savvy individuals often spread their sentiments by advising and influencing others, such as family, friends, and co-workers. Therefore, being inclusive toward the practices and preferences of technically savvy users could help early detection of potential problems.

Some of the expressed concerns, such as user experience, could be addressed via design improvements. On the other hand, other aspects, such as targeted advertising, data sharing, etc., will most likely require improved public policy and regulations along with corresponding enforcement mechanisms. The European Union's (EU) General Data Protection Regulation (GDPR) takes a step in that direction but applies only to EU citizens.

Our findings make a case for explicit design attention to overcoming negative sentiment, even if that means supporting practices that may reduce usage times. Such an approach may help focus attention on use *quality* rather than *quantity*, thus helping reduce negative impacts and, consequently, total abandonment of the system.

## 7 LIMITATIONS

A few limitations must be kept in mind when considering the generalizability of our findings. Our analysis covered a random sample of only 3,000 comments from the complete corpus of comments we scraped. As such, the sample may not necessarily be representative of the entire corpus. Moreover, the relative proportions of the various facets of negative sentiment and non-use within the larger corpus may differ from those within our sample. Even though the comments are naturalistic and unsolicited, they are provided in reaction to a post related to Facebook or the comments of others on the original post. In recent years, a number of these posts have tended to point to news stories that cover Facebook scandals and controversies, which may have led to a larger amount of negativity in the views expressed by the comments.

Based on the posts and discussions on our two chosen sites, we can be reasonably confident that our commenters are technically savvy as a collective. That said, we do not have explicit measures of technical expertise and skills of individual commenters. Moreover, Slashdot and Schneier on

Security receive comments from people all over the world. We do not know the extent to which our findings may be impacted by such cultural variation. Given that a majority of the commenters on both sites appear to be based in the US, we estimate that the impact is likely to be minimal.

## 8 FUTURE WORK

Our findings point to a number of avenues for future work. Our qualitative analyses could serve as training data for machine learning classifiers that utilize Natural Language Processing (NLP) approaches, such as sentiment detection, topic modeling, clustering, opinion mining, etc., thus supporting automation and reducing human effort in analyzing the data. These approaches could be applied at scale to analyze the entire large corpus of comments in order to verify the extent to which findings from the qualitative analyses hold true for the whole corpus. Such analyses can be complemented by data visualization techniques that could help surface further insight, such as notable temporal or topical trends and patterns within the comments.

The influence of technical knowledge on perceptions and practices pertaining to Facebook could be further unpacked by analyzing comments from other technology oriented sites, such as Ars Technica, Wired, etc., and by soliciting input from individuals who are known to possess technical expertise.

## 9 CONCLUSION

Although Facebook is the most widely used social media platform across the world, recent sociopolitical events have served to draw increasing attention to its negative aspects and contributed to calls for abandoning and limiting the use of the service. Our analysis of naturalistic user comments on Facebook related blog posts reveals that technically savvy individuals are especially attentive to these negative aspects since several of the matters are linked to the technological details of Facebook operation. While these individuals share many of the negative views and corresponding non-use practices of the general population, the intensity of these views may be higher, as evidenced by the large proportions of comments that expressed negative sentiment and indicated non-use. Unlike the general population, we rarely found our technically savvy commenters reverting back to using Facebook after abandoning it. As technically savvy individuals are often leading indicators of technology use in the future, we argue that paying attention to their concerns could help designers address issues before they spread to the broader user population. Our findings further demonstrate that analyzing user generated content could be a useful method for quick insight on usage preferences and practices.

## ACKNOWLEDGMENTS

We would like to thank Cavin De Souza for help with the data collection and other supporting tasks. We are grateful to the anonymous reviewers for their insightful comments that helped shape the final version of the paper. We thank Morgan Brockman and Jacob Abbott for proofreading. This research is partially supported by a Faculty Research Award by Google to study technology non-use. The contents of the paper are solely the work of the authors and do not represent the views of the sponsors.

## REFERENCES

- [1] Alessandro Acquisti and Ralph Gross. 2006. Imagined Communities: Awareness, Information Sharing, and Privacy on the Facebook. In *Privacy Enhancing Technologies*, George Danezis and Philippe Golle (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 36–58.
- [2] Icek Ajzen. 1991. The theory of planned behavior. *Organizational Behavior and Human Decision Processes* 50, 2 (1991), 179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- [3] Yeslam Al-Saggaf and Peter Simmons. 2015. Social media in Saudi Arabia: Exploring its use during two natural disasters. *Technological Forecasting and Social Change* 95 (2015), 3–15. <https://doi.org/10.1016/j.techfore.2014.08.013>
- [4] Hunt Allcott and Matthew Gentzkow. 2017. Social Media and Fake News in the 2016 Election. *Journal of Economic Perspectives* 31, 2 (May 2017), 211–36. <https://doi.org/10.1257/jep.31.2.211>
- [5] Helmut Appel, Alexander L. Gerlach, and Jan Crusius. 2016. The interplay between Facebook use, social comparison, envy, and depression. *Current Opinion in Psychology* 9 (2016), 44–49. <https://doi.org/10.1016/j.copsyc.2015.10.006>
- [6] Rosland K. Baker and Katherine M. White. 2011. In Their Own Words: Why Teenagers Don’t Use Social Networking Sites. *Cyberpsychology, Behavior, and Social Networking* 14, 6 (2011), 395–398. <https://doi.org/10.1089/cyber.2010.0016> PMID: 21117974
- [7] Andrew O. Baoill. 2000. Slashdot and the public sphere. *First Monday* 5, 9 (2000). <https://doi.org/10.5210/fm.v5i9.790>
- [8] Eric Baumer, Morgan Ames, Jenna Burrell, Jed Brubaker, and Paul Dourish. 2015. Why study technology non-use? *First Monday* 20, 11 (2015). <https://doi.org/10.5210/fm.v20i11.6310>
- [9] Eric P. S. Baumer. 2018. Socioeconomic Inequalities in the Non Use of Facebook. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI ‘18)*. ACM, New York, NY, USA, Article 616, 14 pages. <https://doi.org/10.1145/3173574.3174190>
- [10] Eric P. S. Baumer, Phil Adams, Vera D. Khovanskaya, Tony C. Liao, Madeline E. Smith, Victoria Schwanda Sosik, and Kaiton Williams. 2013. Limiting, Leaving, and (Re)Lapsing: An Exploration of Facebook Non-use Practices and Experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ‘13)*. ACM, New York, NY, USA, 3257–3266. <https://doi.org/10.1145/2470654.2466446>
- [11] Eric P. S. Baumer, Morgan G. Ames, Jed R. Brubaker, Jenna Burrell, and Paul Dourish. 2014. Refusing, Limiting, Departing: Why We Should Study Technology Non-use. In *CHI ‘14 Extended Abstracts on Human Factors in Computing Systems (CHI EA ‘14)*. ACM, New York, NY, USA, 65–68. <https://doi.org/10.1145/2559206.2559224>
- [12] Eric P. S. Baumer, Jenna Burrell, Morgan G. Ames, Jed R. Brubaker, and Paul Dourish. 2015. On the Importance and Implications of Studying Technology Non-use. *Interactions* 22, 2 (Feb. 2015), 52–56. <https://doi.org/10.1145/2723667>
- [13] Eric P. S. Baumer, Shion Guha, Emily Quan, David Mimno, and Geri K. Gay. 2015. Missing Photos, Suffering Withdrawal, or Finding Freedom? How Experiences of Social Media Non-Use Influence the Likelihood of Reversion. *Social Media + Society* 1, 2 (2015), 2056305115614851. <https://doi.org/10.1177/2056305115614851>
- [14] John Bowers. 1994. The Work to Make a Network Work: Studying CSCW in Action. In *Proceedings of the 1994 ACM Conference on Computer Supported Cooperative Work (CSCW ‘94)*. ACM, New York, NY, USA, 287–298. <https://doi.org/10.1145/192844.193030>
- [15] Nicholas Brody. 2018. Opting Out of Social Media: Online Communication Attitudes Mediate the Relationship Between Personality Factors and Facebook Non-Use. *Southern Communication Journal* 83, 2 (2018), 75–88. <https://doi.org/10.1080/1041794X.2017.1413415>
- [16] Moira Burke and Robert Kraut. 2013. Using Facebook After Losing a Job: Differential Benefits of Strong and Weak Ties. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work (CSCW ‘13)*. ACM, New York, NY, USA, 1419–1430. <https://doi.org/10.1145/2441776.2441936>
- [17] Carole Cadwalladr. 2017. The great British Brexit robbery: How our democracy was hijacked. <https://www.theguardian.com/technology/2017/may/07/the-great-british-brexit-robbery-hijacked-democracy>.
- [18] Jennie Carroll, Steve Howard, Frank Vetere, Jane Peck, and John Murphy. 2002. Just what do the youth of today want? Technology appropriation by young people. In *Proceedings of the 35th Annual Hawaii International Conference on System Sciences*. 1777–1785. <https://doi.org/10.1109/HICSS.2002.994089>
- [19] Kasey Chaulk and Tim Jones. 2011. Online Obsessive Relational Intrusion: Further Concerns About Facebook. *Journal of Family Violence* 26, 4 (01 May 2011), 245–254. <https://doi.org/10.1007/s10896-011-9360-x>
- [20] Fred D. Davis. 1989. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. <http://www.jstor.org/stable/249008>. *MIS Quarterly* 13, 3 (1989), 319–340.
- [21] Claire B. Draucker, Donna S. Martsof, Ratchneewan Ross, and Thomas B. Rusk. 2007. Theoretical Sampling and Category Development in Grounded Theory. *Qualitative Health Research* 17, 8 (2007), 1137–1148. <https://doi.org/10.1177/1049732307308450> PMID: 17928484
- [22] Nicole B. Ellison, Charles Steinfield, and Cliff Lampe. 2007. The Benefits of Facebook “Friends”: Social Capital and College Students’ Use of Online Social Network Sites. *Journal of Computer-Mediated Communication* 12, 4 (2007), 1143–1168. <https://doi.org/10.1111/j.1083-6101.2007.00367.x>
- [23] Kevin Gallagher, Sameer Patil, and Nasir Memon. 2017. New Me: Understanding Expert and Non-Expert Perceptions and Usage of the Tor Anonymity Network. <https://www.usenix.org/conference/soups2017/technical-sessions/presentation/gallagher>. In *Thirteenth Symposium on Usable Privacy and Security (SOUPS 2017)*. USENIX Association, Santa Clara, CA, 385–398.
- [24] Barney G. Glaser and Anselm L. Strauss. 2009. *The discovery of grounded theory: Strategies for qualitative research*. Transaction Publishers.
- [25] Jonathan Grudin. 1988. Why CSCW Applications Fail: Problems in the Design and Evaluation of Organizational Interfaces. In *Proceedings of the 1988 ACM Conference on Computer-supported Cooperative Work (CSCW ‘88)*. ACM, New York, NY, USA, 85–93. <https://doi.org/10.1145/62266.62273>
- [26] Alexander M. Campbell Halavais. 2001. *The Slashdot effect: Analysis of a large-scale public conversation on the World Wide Web*. Ph.D. Dissertation. University of Washington.
- [27] Eszter Hargittai. 2007. Whose Space? Differences Among Users and Non-Users of Social Network Sites. *Journal of Computer-Mediated Communication* 13, 1 (Oct. 2007), 276–297. <https://doi.org/10.1111/j.1083-6101.2007.00396.x>

- [28] Nicholas Hookway. 2008. ‘Entering the blogosphere’: Some strategies for using blogs in social research. *Qualitative Research* 8, 1 (2008), 91–113. <https://doi.org/10.1177/1468794107085298>
- [29] Tiffany Hsu. 2018. For Many Facebook Users, a ‘Last Straw’ That Led Them to Quit. <https://www.nytimes.com/2018/03/21/technology/users-abandon-facebook.html>.
- [30] Iulia Ion, Rob Reeder, and Sunny Consolvo. 2015. “...No one Can Hack My Mind”: Comparing Expert and Non-Expert Security Practices. <https://www.usenix.org/conference/soups2015/proceedings/presentation/ion>. In *Eleventh Symposium On Usable Privacy and Security (SOUPS 2015)*. USENIX Association, Ottawa, 327–346.
- [31] A. K. M. Najmul Islam and Sameer Patil. 2015. Engagement and Well-being on Social Network Sites. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW ‘15)*. ACM, New York, NY, USA, 375–382. <https://doi.org/10.1145/2675133.2675299>
- [32] Hélène Joffe and Lucy Yardley. 2004. Content and thematic analysis. In *Research methods for clinical and health psychology*, David F. Marks and Lucy Yardley (Eds.). Sage, 56–68.
- [33] Adam N. Joinson. 2008. Looking at, Looking Up or Keeping Up with People?: Motives and Use of Facebook. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ‘08)*. ACM, New York, NY, USA, 1027–1036. <https://doi.org/10.1145/1357054.1357213>
- [34] Hee-Woong Kim and Atreyi Kankanhalli. 2009. Investigating User Resistance to Information Systems Implementation: A Status Quo Bias Perspective. <http://www.jstor.org/stable/20650309>. *MIS Quarterly* 33, 3 (2009), 567–582.
- [35] Ksenia Koroleva, Hanna Krasnova, Natasha Veltri, and Oliver Günther. 2011. It’s All About Networking! Empirical Investigation of Social Capital Formation on Social Network Sites. <https://boris.unibe.ch/id/eprint/47120>. In *Proceedings of the 32nd International Conference on Information Systems (ICIS 2011)*. <https://doi.org/10.7892/boris.47120>
- [36] Hanna Krasnova, Helena Wenninger, Thomas Widjaja, and Peter Buxmann. 2013. Envy on Facebook: A Hidden Threat to Users’ Life Satisfaction?. <https://boris.unibe.ch/id/eprint/47080>. In *Proceedings of the 11th International Conference on Wirtschaftsinformatik (WI) / Business Information Systems (WI 2013)*. <https://doi.org/10.7892/boris.47080>
- [37] Jérôme Kunegis, Andreas Lommatzsch, and Christian Bauckhage. 2009. The Slashdot Zoo: Mining a Social Network with Negative Edges. In *Proceedings of the 18th International Conference on World Wide Web (WWW ‘09)*. ACM, New York, NY, USA, 741–750. <https://doi.org/10.1145/1526709.1526809>
- [38] Grace Chi En Kwan and Marko M. Skoric. 2013. Facebook bullying: An extension of battles in school. *Computers in Human Behavior* 29, 1 (2013), 16–25. <https://doi.org/10.1016/j.chb.2012.07.014>
- [39] Cliff Lampe and Paul Resnick. 2004. Slash(Dot) and Burn: Distributed Moderation in a Large Online Conversation Space. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ‘04)*. ACM, New York, NY, USA, 543–550. <https://doi.org/10.1145/985692.985761>
- [40] Cliff Lampe, Jessica Vitak, and Nicole Ellison. 2013. Users and Nonusers: Interactions Between Levels of Adoption and Social Capital. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work (CSCW ‘13)*. ACM, New York, NY, USA, 809–820. <https://doi.org/10.1145/2441776.2441867>
- [41] Cliff A. C. Lampe, Erik Johnston, and Paul Resnick. 2007. Follow the Reader: Filtering Comments on Slashdot. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ‘07)*. ACM, New York, NY, USA, 1253–1262. <https://doi.org/10.1145/1240624.1240815>
- [42] Liette Lapointe and Suzanne Rivard. 2005. A Multilevel Model of Resistance to Information Technology Implementation. <http://www.jstor.org/stable/25148692>. *MIS Quarterly* 29, 3 (2005), 461–491.
- [43] Issie Lapowsky. 2018. The 21 (and Counting) Biggest Facebook Scandals of 2018. <https://www.wired.com/story/facebook-scandals-2018/>.
- [44] Issie Lapowsky. 2018. Mark Zuckerberg Answers to Congress For Facebook’s Troubles. <https://www.wired.com/story/mark-zuckerberg-congress-facebook-troubles/>.
- [45] Nikolina Ljepava, R. Robert Orr, Sean Locke, and Craig Ross. 2013. Personality and social characteristics of Facebook non-users and frequent users. *Computers in Human Behavior* 29, 4 (2013), 1602–1607. <https://doi.org/10.1016/j.chb.2013.01.026>
- [46] G. M. Marakas and S. Hornik. 1996. Passive resistance misuse: Overt support and covert recalcitrance in IS implementation. *European Journal of Information Systems* 5, 3 (01 September 1996), 208–219. <https://doi.org/10.1057/ejis.1996.26>
- [47] Megan A. Moreno, Lauren A. Jelenchick, Katie G. Egan, Elizabeth Cox, Henry Young, Kerry E. Gannon, and Tara Becker. 2011. Feeling bad on Facebook: Depression disclosures by college students on a social networking site. *Depression and Anxiety* 28, 6 (2011), 447–455. <https://doi.org/10.1002/da.20805>
- [48] Robin L. Nabi, Abby Prestin, and Jiyeon So. 2013. Facebook Friends with (Health) Benefits? Exploring Social Network Site Use and Perceptions of Social Support, Stress, and Well-Being. *Cyberpsychology, Behavior, and Social Networking* 16, 10 (2013), 721–727. <https://doi.org/10.1089/cyber.2012.0521> PMID: 23790356.
- [49] Blair Nonnecke and Jenny Preece. 2001. Why Lurkers Lurk. <https://aisel.aisnet.org/amcis2001/294>. In *Proceedings of the Americas Conference on Information Systems (AMCIS 2001)*.
- [50] Wanda J. Orlikowski. 1993. Learning from notes: Organizational issues in groupware implementation. *The Information Society* 9, 3 (1993), 237–250. <https://doi.org/10.1080/01972243.1993.9960143>
- [51] Nathaniel Poor. 2005. Mechanisms of an Online Public Sphere: the Website Slashdot. *Journal of Computer-Mediated Communication* 10, 2 (2005), JCMC1028. <https://doi.org/10.1111/j.1083-6101.2005.tb00241.x>
- [52] Kenneth Portier, Greta E. Greer, Lior Rokach, Nir Ofek, Yafei Wang, Prakhari Biyani, Mo Yu, Siddhartha Banerjee, Kang Zhao, Prasennjit Mitra, and John Yen. 2013. Understanding Topics and Sentiment in an Online Cancer Survivor Community. *JNCI Monographs* 2013, 47 (2013), 195–198. <https://doi.org/10.1093/jncimonographs/ltg025>
- [53] Jenny Preece and Diane Maloney-Krichmar. 2005. Online Communities: Design, Theory, and Practice. *Journal of Computer-Mediated Communication* 10, 4 (July 2005). <https://doi.org/10.1111/j.1083-6101.2005.tb00264.x>
- [54] Yan Qu, Philip Fei Wu, and Xiaoqing Wang. 2009. Online Community Response to Major Disaster: A Study of Tianya Forum in the 2008 Sichuan Earthquake. In *Proceedings of the 42nd Hawaii International Conference on System Sciences (HICSS 2009)*. 1–11. <https://doi.org/10.1109/HICSS.2009.330>
- [55] Lee Rainie, Aaron Smith, and Maeve Duggan. 2013. Coming and going on Facebook. <http://pewinternet.org/Reports/2013/Coming-and-going-on-facebook.aspx>. *Pew Research Center’s Internet & American Life Project* (February 2013).
- [56] Tracie Ryan and Sophia Xenos. 2011. Who uses Facebook? An investigation into the relationship between the Big Five, shyness, narcissism, loneliness, and Facebook usage. *Computers in Human Behavior* 27, 5 (2011), 1658–1664. <https://doi.org/10.1016/j.chb.2011.02.004>
- [57] Christine Satchell and Paul Dourish. 2009. Beyond the User: Use and Non-use in HCI. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group: Design: Open 24/7 (OZCHI ‘09)*. ACM, New York, NY, USA, 9–16. <https://doi.org/10.1145/1526709.1526809>

- //doi.org/10.1145/1738826.1738829
- [58] Sarita Yardi Schoenebeck. 2014. Giving Up Twitter for Lent: How and Why We Take Breaks from Social Media. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. ACM, New York, NY, USA, 773–782. <https://doi.org/10.1145/2556288.2556983>
  - [59] Neil Selwyn. 2003. Apart from technology: Understanding people's non-use of information and communication technologies in everyday life. *Technology in Society* 25, 1 (2003), 99–116. [https://doi.org/10.1016/S0160-791X\(02\)00062-3](https://doi.org/10.1016/S0160-791X(02)00062-3)
  - [60] Neil Selwyn, Stephen Gorard, and John Furlong. 2005. Whose Internet is it Anyway?: Exploring Adults' (Non)Use of the Internet in Everyday Life. *European Journal of Communication* 20, 1 (2005), 5–26. <https://doi.org/10.1177/0267323105049631>
  - [61] Holly B. Shakya and Nicholas A. Christakis. 2017. A New, More Rigorous Study Confirms: The More You Use Facebook, the Worse You Feel. <https://hbr.org/2017/04/a-new-more-rigorous-study-confirms-the-more-you-use-facebook-the-worse-you-feel>.
  - [62] Viswanath Venkatesh and Hillol Bala. 2008. Technology Acceptance Model 3 and a Research Agenda on Interventions. *Decision Sciences* 39, 2 (2008), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>
  - [63] Viswanath Venkatesh, Michael G. Morris, Gordon B. Davis, and Fred D. Davis. 2003. User Acceptance of Information Technology: Toward a Unified View. <http://www.jstor.org/stable/30036540>. *MIS Quarterly* 27, 3 (2003), 425–478.
  - [64] Pieter Verdegem and Pascal Verhoest. 2009. Profiling the non-user: Rethinking policy initiatives stimulating ICT acceptance. *Telecommunications Policy* 33, 10 (2009), 642–652. <https://doi.org/10.1016/j.telpol.2009.08.009>
  - [65] Jessica Vitak, Cliff Lampe, Rebecca Gray, and Nicole B. Ellison. 2012. “Why Won’t You Be My Facebook Friend?”: Strategies for Managing Context Collapse in the Workplace. In *Proceedings of the 2012 iConference (iConference '12)*. ACM, New York, NY, USA, 555–557. <https://doi.org/10.1145/2132176.2132286>
  - [66] Pamela Wisniewski, A. K. M. Najmul Islam, Bart P. Knijnenburg, and Sameer Patil. 2015. Give Social Network Users the Privacy They Want. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)*. ACM, New York, NY, USA, 1427–1441. <https://doi.org/10.1145/2675133.2675256>
  - [67] Sally Wyatt. 2003. Non-users also matter: The construction of users and non-users of the Internet. In *Now Users Matter: The Co-construction of Users and Technology*, Nelly Oudshoorn and Trevor Pinch (Eds.). MIT Press, 67–79.
  - [68] Susan Wyche and Eric P. S. Baumer. 2017. Imagined Facebook: An exploratory study of non-users’ perceptions of social media in Rural Zambia. *New Media & Society* 19, 7 (2017), 1092–1108. <https://doi.org/10.1177/1461444815625948>